CHUKA UNIVERSITY

INSTI NAL DATA SCIENCE FOR STATISTICIANS

ASSIGNMENT TWO

SENETWICKS

Kimathi Kelvin

Email: kelvin.kimathi@students.chuka.ac.ke

QUESTION 1 a) Describe four relevant variables in this study Data climate and sea-sure-economics.For each variable, state

Climate Variables

1. Percent Attachment

Variable Description

Percent attachment with coastal protection wetlands

Data source

Data collected from NOAA Coastal Services Center, provides science-based decision-support tools and information

Limitations

- Some Metropolitan Statistical Areas (coastal) aggregated, unknown if underestimate, relying on aggregate or other measurements. Some areas may also have measurements within more detailed scales.

Relevance

- Coastal ecosystems offer ecosystem services such as protection from storms, carbon sequestration, and habitat for fish and wildlife. Wetlands can reduce storm damage by absorbing and dissipating wave energy, trapping sediment, and providing habitat that helps stabilize shorelines. Measuring attachment helps assess how well these ecosystems are maintaining their role in coastal resilience.

2. Wave Temperature

Unit of measurement

Degrees Celsius (°C) or degrees Fahrenheit (°F)

Data source

Data collected from buoys maintained by NOAA's National Data Buoy Center

Limitations

- Wave buoys are not the most dense data source, but important for calibration, in combination with other satellite or other data collected coastal areas. Data report differs may use a different measurement or a standard calculation.

Relevance

- Ocean surface temperature affects coastal weather patterns, storm intensity, and marine ecosystems. Warm water can fuel stronger storms and affect the distribution of marine species. Monitoring wave temperature helps track climate change impacts and predict storm behavior.

Socioeconomic Variables

3. Poverty Rate/Household Income Level

Variable Description

Percentage of households below poverty line or median household income relative to regional averages

Data source

U.S. Census Bureau, American Community Survey (ACS) 5-year estimates

Limitations

- Data is collected at county or tract level, not individual household level, so it lacks granularity. Recent changes in methodology may affect comparability across years.

Relevance

- Coastal communities with lower income may have fewer resources for storm preparedness, evacuation, and post-disaster recovery. Understanding poverty distribution helps target aid and assess which communities are most vulnerable to economic impacts from climate change.

Sea-to-address 4

Unit of measurement

Number of coastal flood events, or percentage of coastal flood events

Data source

NOAA Coastal Services Center's Sea-to-address Tool, combines tide gauge data, satellite imagery, and coastal elevation models to map flood extents and frequency

Limitations

- Historical data may be incomplete for some locations, and future projections depend on sea level rise models that have uncertainties.

Relevance

- Coastal flooding is a direct consequence of climate change (sea level rise, storm surge) and affects both ecosystems and human communities. Tracking flood events helps assess which areas are most at risk and how conditions are changing over time.

5. Education Level (Literacy Rate)

Unit of measurement

Percentage of adults aged 25+ with a high school diploma or higher

Data source

U.S. Census Bureau, American Community Survey (ACS) 5-year estimates

Limitations

- Education data may lag behind current workforce skills and community capacity changes. Self-reported literacy rates may differ from actual educational attainment. Data may not be available for all coastal regions.

Relevance

- Education levels affect a community's ability to adapt to climate change, access information about risks and preparedness, and participate in decision-making processes. Higher literacy can enable more effective responses to coastal hazards.

QUESTION 2

The team must combine multiple datasets into a unified analysis. Explain the concepts of

Data Integrating, Data Wrangling, Data Munging, Data Merging, Data Mapping, Data Joining, Data Linking, Data Blending, Data Aggregation, Data Scraping, Data Ingestion, Data Extraction, Data Parsing, Data Cleaning, Data Validation, Data Enrichment, Data Transformation, Data Normalization, Data Standardization, Data Deduplication, Data Quality Assurance, Data Profiling, Data Sampling, Data Partitioning, Data Stratification, Data Imputation, Data Encoding, Data Decoding, Data Compression, Data Decompression, Data Encryption, Data Decryption, Data Hashing, Data Tokenization, Data Masking, Data Anonymization, Data Pseudonymization, Data Aggregation, Data Disaggregation, Data Pivoting, Data Unpivoting, Data Reshaping, Data Transposition, Data Filtering, Data Sorting, Data Ordering, Data Indexing, Data Cataloging, Data Tagging, Data Annotation, Data Labeling, Data Classification, Data Clustering, Data Segmentation, Data Partitioning, Data Stratification, Data Imputation, Data Encoding, Data Decoding, Data Compression, Data Decompression, Data Encryption, Data Decryption, Data Hashing, Data Tokenization, Data Masking, Data Anonymization, Data Pseudonymization, Data Aggregation, Data Disaggregation, Data Pivoting, Data Unpivoting, Data Reshaping, Data Transposition, Data Filtering, Data Sorting, Data Ordering, Data Indexing, Data Cataloging, Data Tagging, Data Annotation, Data Labeling, Data Classification, Data Clustering, Data Segmentation

1. Data Joining/Merging

Definition

Combining two or more datasets by matching records based on shared keys or common columns, creating a wider dataset when datasets share attributes but contain different records or measures.

Real-world

Combining weather station data with coastal flood records

Why it matters

Allows you to see relationships between variables that exist in different sources, enabling more comprehensive analysis than any single dataset alone.

Types

- Inner join: keeps only records that match in both datasets
- Left join: keeps all records from the first dataset
- Outer join: keeps all records from both datasets

Key issue

Handling mismatches when some records don't have matches in the other dataset, creating missing data in the combined dataset.

2. Data Wrangling

Definition

Transforming raw data into a clean, structured format for analysis, including cleaning, reshaping, and enriching data from various sources.

Real-world

Converting messy sensor logs from different weather stations into a standardized format with consistent units and timestamps.

Why it matters

Raw data is rarely in the format needed for analysis; wrangling makes it usable.

Key steps

- Parsing dates from different formats
- Standardizing units (Celsius to Fahrenheit)
- Handling missing values
- Removing duplicate records

3. Data Munging

Definition

Manually fixing data errors, inconsistencies, and formatting problems that prevent automated processing.

Real-world

Correcting station names that have typos or different spellings across datasets.

Why it matters

Automated tools can't handle everything; munging addresses the edge cases that require human judgment.

Contrasted with

Wrangling (more systematic, less ad-hoc)

4. Data Merging

Definition

Combining datasets by stacking them vertically (adding more rows) or horizontally (adding more columns).

Real-world

Stacking monthly weather reports into a yearly dataset.

Why it matters

Different merge operations serve different purposes—stacking increases sample size, joining adds new variables.

Contrasted with

Joining (specifically combines by matching keys)

5. Data Mapping

Definition

Creating a visual representation of data to identify patterns, relationships, or geographic distributions.

Real-world

Plotting storm locations on a map to see which coastal areas are most affected.

Why it matters

Visualization reveals patterns that tables of numbers don't show.

Types

- Geographic mapping
- Heat maps
- Scatter plots

6. Data Aggregation

Definition

Summarizing detailed data into higher-level summaries (totals, averages, counts).

Real-world

Calculating monthly average temperatures from daily readings.

Why it matters

Reduces data volume while preserving key patterns; enables comparisons across time periods or regions.

Trade-off

Loses detail in exchange for clarity.

7. Data Ingestion

Definition

Importing data from various sources into a system where it can be stored and processed.

Real-world

Loading CSV files, API responses, and database exports into a central repository.

Why it matters

The first step in any data pipeline; ensures all data enters the system in a consistent format.

Sources

- Files (CSV, Excel, JSON)
- APIs
- Databases
- Sensors

8. Data Extraction

Definition

Pulling specific information from unstructured or semi-structured sources.

Real-world

Extracting temperature readings from PDF weather reports or web pages.

Why it matters

Much valuable data isn't in clean, structured formats—extraction makes it usable.

Challenges

- Changing formats
- Inconsistent structure
- Embedded in narrative text

9. Data Parsing

Definition

Breaking structured text into components according to rules or grammar.

Real-world

Reading date strings like "2024-03-15" or "March 15, 2024" and recognizing the year, month, and day.

Why it matters

Computers need explicit instructions to interpret text as meaningful data.

Common formats

- CSV
- JSON
- XML
- Log files

10. Data Cleaning

Definition

Fixing errors, inconsistencies, and missing values in the dataset.

Real-world

Correcting station names that have typos or different spellings across datasets.

Why it matters

Dirty data leads to wrong conclusions; cleaning ensures reliable results.

Common tasks

- Handling missing values
- Removing duplicates
- Correcting typos
- Standardizing formats

11. Data Validation

Definition

Checking that data meets expected rules or constraints.

Real-world

Ensuring temperatures are within physically possible ranges.

Why it matters

Catches errors before they propagate through analysis.

Types

- Range checks
- Format checks
- Consistency checks

12. Data Enrichment

Definition

Adding new information to existing records from other sources.

Real-world

Adding elevation data to weather stations based on their coordinates.

Why it matters

Enriched data enables more sophisticated analysis.

Example

Combining station locations with topographic maps to calculate distance from coast.

13. Data Standardization

Definition

Converting data to consistent units, formats, and conventions.

Real-world

Converting all temperatures to Celsius, all dates to ISO format (YYYY-MM-DD).

Why it matters

Inconsistent units make comparisons impossible.

Common standards

- Metric units
- ISO formats
- UTF-8 encoding

14. Data Deduplication

Definition

Identifying and removing duplicate records.

Real-world

Removing the same storm event recorded by two different stations.

Why it matters

Duplicates inflate counts and bias analysis.

Approaches

- Exact matching
- Fuzzy matching (for near-duplicates)

15. Data Quality Assurance

Definition

Overall process of ensuring data is accurate, complete, and reliable.

Real-world

Running automated checks before publishing analysis.

Why it matters

Stakeholders need confidence in the data before making decisions.

Components

- Validation
- Monitoring
- Documentation

16. Data Profiling

Definition

Analyzing data to understand its structure, patterns, and quality before processing.

Real-world

Checking what percentage of values are missing in each column before starting analysis.

Why it matters

Helps identify problems early and informs cleaning strategies.

Outputs

- Summary statistics
- Frequency distributions
- Missing value reports

17. Data Sampling

Definition

Selecting a representative subset of data for analysis.

Real-world

Using 10% of historical weather data for initial exploratory analysis.

Why it matters

Enables faster iteration when working with large datasets.

Methods

- Random sampling
- Stratified sampling
- Time-based sampling

18. Data Partitioning

Definition

Splitting data into separate groups for training, validation, and testing.

Real-world

Using 2015-2020 data for training, 2021 for validation, 2022-2024 for testing.

Why it matters

Prevents overfitting and enables honest evaluation of predictive models.

Common splits

- 70% training
- 15% validation
- 15% test

19. Data Stratification

Definition

Ensuring proportional representation of different groups across partitions.

Real-world

Making sure each region's weather patterns appear in training, validation, and test sets in the same proportions as the full dataset.

Why it matters

Prevents sampling bias that could skew results.

20. Data Imputation

Definition

Filling in missing values with estimated values.

Real-world

Using average temperature from nearby stations to fill gaps when one station's data is missing.

Why it matters

Many analysis methods can't handle missing data; imputation allows them to proceed.

Methods

- Mean/median imputation
- Forward fill (carry last observation forward)
- Interpolation
- Model-based prediction

21. Data Encoding

Definition

Converting categorical data into numerical representations.

Real-world

Converting station names into numbers for machine learning algorithms.

Why it matters

Most algorithms require numerical input.

Types

- One-hot encoding
- Label encoding
- Target encoding

22. Data Decoding

Definition

Converting encoded data back to its original form.

Real-world

Translating predicted station IDs back to station names for interpretation.

Why it matters

Needed to interpret model outputs in meaningful terms.

23. Data Compression

Definition

Reducing data size while preserving information.

Real-world

Storing historical weather data in compressed formats to save disk space.

Why it matters

Enables storage and faster transmission of large datasets.

Types

- Lossless (no information loss)
- Lossy (some information loss acceptable)

24. Data Decompression

Definition

Restoring compressed data to its original size.

Real-world

Unzipping files before analysis.

Why it matters

Required before any processing can begin.

25. Data Encryption

Definition

Protecting data by encoding it so only authorized parties can read it.

Real-world

Securing sensitive location data of endangered species.

Why it matters

Protects privacy and prevents unauthorized access.

Types

- Symmetric encryption
- Asymmetric encryption

26. Data Decryption

Definition

Authorized access to encrypted data.

Real-world

Researchers with permission accessing protected species locations.

Why it matters

Balances security with legitimate access needs.

27. Data Hashing

Definition

Creating a fixed-length fingerprint from data.

Real-world

Verifying that a file hasn't been corrupted during transfer.

Why it matters

Detects changes or corruption in data.

Properties

- Same input always produces same output
- Small changes produce completely different output
- Cannot be reversed

28. Data Tokenization

Definition

Replacing sensitive data with non-sensitive tokens.

Real-world

Replacing exact coordinates with grid cell IDs.

Why it matters

Allows analysis without exposing sensitive information.

29. Data Masking

Definition

Hiding portions of data while preserving its structure.

Real-world

Showing only the last four digits of a phone number.

Why it matters

Allows partial visibility for verification without full exposure.

30. Data Anonymization

Definition

Removing personal identifiers so individuals can't be traced.

Real-world

Removing exact coordinates of endangered species sightings.

Why it matters

Protects privacy while enabling ecological research.

31. Data Pseudonymization

Definition

Replacing identifiers with consistent fake values.

Real-world

Assigning each tracked animal a random ID that stays the same across its records.

Why it matters

Allows tracking patterns over time without knowing real identities.

32. Data Aggregation

Definition

Combining multiple records into summary statistics.

Real-world

Calculating monthly average temperatures from daily readings.

Why it matters

Reduces data volume while preserving key patterns.

33. Data Disaggregation

Definition

Breaking aggregated data into more detailed components.

Real-world

Estimating hourly temperatures from daily averages.

Why it matters

Enables analysis at finer resolution than the original data allows.

34. Data Pivoting

Definition

Reorganizing data from long to wide format.

Real-world

Converting rows of station readings into columns for each station.

Why it matters

Different analyses require different data shapes.

35. Data Unpivoting

Definition

Reorganizing data from wide to long format.

Real-world

Converting station columns into rows with station identifiers.

Why it matters

Many tools prefer long format for analysis.

36. Data Reshaping

Definition

Restructuring data to change its dimensions.

Real-world

Converting time series into a matrix for machine learning.

Why it matters

Different algorithms require different input structures.

37. Data Transposition

Definition

Swapping rows and columns.

Real-world

Flipping a dataset so stations become rows instead of columns.

Why it matters

Simple but frequently needed for compatibility.

38. Data Filtering

Definition

Selecting records that meet specified criteria.

Real-world

Keeping only records from coastal stations.

Why it matters

Focuses analysis on relevant subsets.

39. Data Sorting

Definition

Arranging records in order.

Real-world

Ordering by date before calculating rolling averages.

Why it matters

Many operations assume sorted data.

40. Data Ordering

Definition

Establishing sequence in data.

Real-world

Time series must be ordered chronologically.

Why it matters

Order matters for time-based analysis.

41. Data Indexing

Definition

Creating a lookup structure for faster access.

Real-world

Indexing station IDs for quick retrieval.

Why it matters

Speeds up queries on large datasets.

42. Data Cataloging

Definition

Documenting what data exists and where it's stored.

Real-world

Maintaining a list of all weather datasets with their locations and descriptions.

Why it matters

Helps people find and understand available data.

43. Data Tagging

Definition

Adding labels to data for organization.

Real-world

Tagging records by region or quality level.

Why it matters

Enables filtering and grouping by attributes.

44. Data Annotation

Definition

Adding notes or metadata to records.

Real-world

Marking which records were manually verified.

Why it matters

Provides context that isn't in the raw data.

45. Data Labeling

Definition

Assigning categorical values to records.

Real-world

Labeling storm events as "coastal" or "inland".

Why it matters

Enables supervised machine learning.

46. Data Classification

Definition

Assigning records to predefined categories.

Real-world

Classifying storms by intensity (tropical depression, tropical storm, hurricane).

Why it matters

Enables pattern recognition across different types.

47. Data Clustering

Definition

Grouping similar records without predefined categories.

Real-world

Finding natural groupings of storm tracks.

Why it matters

Reveals patterns you didn't know to look for.

48. Data Segmentation

Definition

Dividing data into meaningful groups.

Real-world

Separating coastal vs. inland stations for different analysis.

Why it matters

Different segments may behave differently.

49. Data Partitioning

Definition

Splitting data into separate groups for training, validation, and testing.

Real-world

Using 2015-2020 data for training, 2021 for validation, 2022-2024 for testing.

Why it matters

Prevents overfitting and enables honest evaluation of predictive models.

50. Data Stratification

Definition

Ensuring proportional representation of different groups across partitions.

Real-world

Making sure each region's weather patterns appear in training, validation, and test sets in the same proportions as the full dataset.

Why it matters

Prevents sampling bias that could skew results.

QUESTION 3

Do not print a line listed in Line 7 of Third part variables.

```
Line 1 Hello world!
Line 2 Hello world!
Line 3 Hello world!
Line 4 Hello world!
Line 5 Hello world!
Line 6 Hello world!
Line 7 Hello world!
Line 8 Hello world!
Line 9 Hello world!
Line 10 Hello world!
```

```
# Create the file with 10 lines
cat > hello.sh << EOF
echo "Line 1 Hello world!"
echo "Line 2 Hello world!"
echo "Line 3 Hello world!"
echo "Line 4 Hello world!"
echo "Line 5 Hello world!"
echo "Line 6 Hello world!"
echo "Line 7 Hello world!"
echo "Line 8 Hello world!"
echo "Line 9 Hello world!"
echo "Line 10 Hello world!"
EOF

# Make it executable
chmod +x hello.sh

# Run it to see the output
./hello.sh

# Now create the third part variables.
cat > variables.sh << EOF
# Third part variables
name="Kimathi"
age=25
city="Nairobi"
EOF

# Make it executable
chmod +x variables.sh

# Run it to load the variables
source variables.sh

# Now print the variables to verify
echo $name
echo $age
echo $city
```